

VDC Teacher Mobile Application

Product Requirements Document (PRD)

Project Overview

Develop a production-ready Android application for VDC teachers to manage attendance, test marks, assignment marks, reports, notes, and automated SMS notifications using a secure cloud backend.

Business Goals

Reduce manual work, digitize academic records, automate attendance and reporting, and provide a scalable foundation for future ERP modules.

Scope

Version 1 includes Teacher Authentication, Dashboard, Attendance, Test & Assignment Marks, Reports, Notes Repository, SMS Integration, Cloud Database, and Android APK deployment.

Technology Stack

Frontend: React Native + Expo + TypeScript + NativeWind + Zustand + TanStack Query.

Backend Platform: Supabase (Authentication, PostgreSQL, Storage).

Database: PostgreSQL.

SMS: Fast2SMS API.

Charts: React Native Chart Kit.

Functional Modules

1. Authentication
2. Dashboard
3. Class Management
4. Attendance Management
5. Test Marks
6. Assignment Marks
7. Reports (Daily, Monthly, Quarterly)
8. Notes Repository
9. SMS Integration

Security

JWT Authentication, HTTPS, Password Hashing, Role-based Authorization, Row Level Security, Audit Logging.

Core Database Entities

Teacher, Student, Class, Subject, TeacherClass, Attendance, Test, TestMarks, Assignment, AssignmentMarks, Notes, SMSLog.

Development Roadmap

Sprint 1: Authentication & Dashboard

Sprint 2: Attendance & SMS

Sprint 3: Test & Assignment

Sprint 4: Reports

Sprint 5: Notes

Sprint 6: Testing & APK Release

Future Enhancements

Admin Portal, Student Portal, Parent Portal, Push Notifications, Fee Management, AI Analytics, OCR Attendance, Voice Attendance.

Definition of Done

Teachers can securely log in, manage attendance and marks, generate reports, upload notes, send SMS to absentees, and use a tested Android APK backed by a secure cloud platform.